

(P)

$$\min_{x \in \Omega} f(x)$$

where $\Omega \subset \mathbb{R}^n$ (often: $\Omega = \mathbb{R}^n$)
 $f: \Omega \rightarrow \mathbb{R}$

Theorem: (First-order necessary conditions) Let $\Omega \subset \mathbb{R}^n$, $f \in C^1(\Omega)$.

If $x^* \in \Omega$ is a local minimum of f on Ω , then for any $d \in \mathbb{R}^n$ that is feasible at x^* , $\nabla f(x^*)^T d \geq 0$.

Theorem (Second-order necessary conditions)

Let $\Omega \subset \mathbb{R}^n$, $f \in C^2(\Omega)$. If $x^* \in \Omega$ is a local minimum of f on Ω , then for any feasible direction

(i) $\nabla f(x^*)^T d \geq 0$

(ii) If $\nabla f(x^*)^T d = 0$, then $d^T \nabla^2 f(x^*) d \geq 0$.

Theorem: (Second-order sufficient conditions)

$f \in C^2(\mathbb{R}^n)$. $x^* \in \mathbb{R}^n$ such that:

(i) $\nabla f(x^*) = 0$

(ii) $\nabla^2 f(x^*) > 0$, i.e. positive definite matrix

Then: x^* is a strict local minimum of f .

Examples: ① $f(x) = x^2$, $f''(x) = 2$, thus satisfies
 SOSC at $x=0 \Rightarrow x=0$ is strict minimum!

② $f(x) = x^4$, $f'(x) = 4x^3$, thus $f''(0) = 0$
 $\Rightarrow$ does not satisfy SOSC, but is still
 strict minimum!

③: $f(x,y) = \frac{1}{2}(x,y) \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} - (3,5) \begin{pmatrix} x \\ y \end{pmatrix}$
 $\Rightarrow \nabla^2 f(x,y) = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$

is positive definite as
 $\det(\dots) = 4 - 1 > 0$,
 $\text{Tr}(\dots) = 4 > 0$.

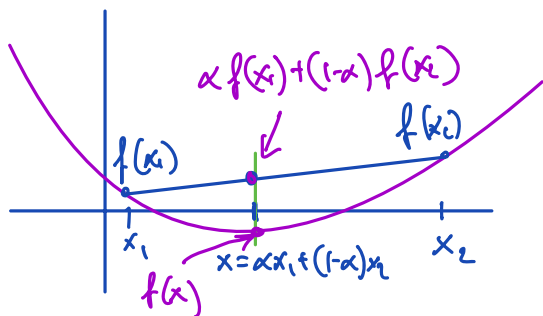
Convexity of functions

When is finding a local minimum sufficient to find the global minimum?

Def: A function f defined on a convex set $\Omega \subset \mathbb{R}^n$ is called convex if for every $x_1, x_2 \in \Omega$ and every $\alpha \in [0, 1]$ it holds that

$$f(\alpha x_1 + (1-\alpha)x_2) \leq \alpha f(x_1) + (1-\alpha)f(x_2)$$

If " $<$ " for every $\alpha \in (0, 1)$, then f is strictly convex.



Def: If $-f$ is convex, we say that f is concave.

Observations: $f_1, f_2: \Omega \rightarrow \mathbb{R}$ convex, then

- $f_1 + f_2$ convex

$$\begin{aligned} [x_1, x_2 \in \Omega \Rightarrow (f_1 + f_2)(\alpha x_1 + (1-\alpha)x_2) &= \\ &= f_1(\alpha x_1 + (1-\alpha)x_2) + f_2(\alpha x_1 + (1-\alpha)x_2) \stackrel{f_1, f_2 \text{ conv.}}{\leq} \\ &\leq \alpha f_1(x_1) + (1-\alpha)f_1(x_2) + \alpha f_2(x_1) + (1-\alpha)f_2(x_2) \\ &= \alpha(f_1 + f_2)(x_1) + (1-\alpha)(f_1 + f_2)(x_2) \quad \checkmark] \end{aligned}$$

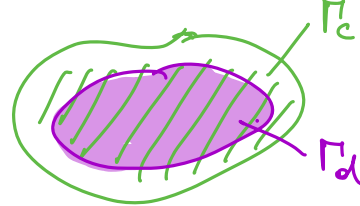
- $a f_1$ convex for $a \geq 0$. $\checkmark$

- Let f be convex on Ω . The set $\Gamma_c = \{x \in \Omega \mid f(x) \leq c\}$ is convex for all $c \in \mathbb{R}$.
"level set"

[For $x_1, x_2 \in \Gamma_c$, $f(x_1), f(x_2) \leq c$

$$\Rightarrow f(\alpha x_1 + (1-\alpha)x_2) \leq \alpha f(x_1) + (1-\alpha)f(x_2) \leq \alpha c + (1-\alpha)c = c$$

$\Rightarrow \Gamma_c$ convex.]



Convexity and differentiability:

Theorem: Let $f \in C^1(\Omega)$. Then f is convex if and only if

$$f(y) \geq f(x) + \nabla f(x)^T (y-x) \quad \text{for all } x, y \in \Omega.$$

"gradient inequality".

Proof: " $\Rightarrow$ ": f convex $\Rightarrow$ for all $\alpha \in [0,1]$:

$$f(\alpha y + (1-\alpha)x) \leq \alpha f(y) + (1-\alpha)f(x)$$

$$\alpha \neq 0: \frac{f(\alpha y + (1-\alpha)x) - f(x)}{\alpha} \leq f(y) - f(x)$$

$$\alpha \rightarrow 0: \nabla f(x)^T (y-x) \leq f(y) - f(x) \quad \checkmark$$

" $\Leftarrow$ ": Assume gradient inequality holds, $x_1, x_2 \in \Omega$

$$(i) \quad f(x_1) \geq f(x) + \nabla f(x)^T (x_1 - x)$$

$$(ii) \quad f(x_2) \geq f(x) + \nabla f(x)^T (x_2 - x)$$

$\alpha(i) + (1-\alpha)(ii)$

$$\alpha f(x_1) + (1-\alpha)f(x_2) \geq f(x) + \nabla f(x)^T (\alpha x_1 + (1-\alpha)x_2 - x)$$

Choose: $x := \alpha x_1 + (1-\alpha)x_2 \Rightarrow$

$$f(\alpha x_1 + (1-\alpha)x_2) \leq \alpha f(x_1) + (1-\alpha)f(x_2)$$

□.

